METRIC

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PERFORMANCE SPECIFICATION

SPEED PICKUP, MAGNETIC (METRIC)

This specification is approved for use within U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance and inspection requirements for magnetic speed pickups hereinafter referred to as pickup (see 6.1).

1.2 Classification. The pickups covered by this specification shall be of the following types as specified (see 6.2).

TYPE I	High pressure compressor speed (N_H) in accordance with drawing 12286861.
TYPE II	Power turbine speed (N_P) in accordance with drawing 12286840.

Beneficial comments (recommendations, additions,. deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC 2990

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

P-C-437 - Cleaning Compound, High Pressure (Steam Cleaner).

DEPARTMENT OF DEFENSE

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-810 - Environmental Test Methods.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12302067 - Adaptor, Speedometer - Tachometer.
12302350 - Speed Pickup, Electromagnetic.

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12286861 - High Pressure Compressor Speed (N_H).
12286840 - Power Turbine Speed (N_P).

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC/ANSI ZI.4 - Sampling Procedures and Tables for Inspections by Attributes.

(Application for copies should be addressed to The Society for Quality Control, P.O. Box 305, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for pick-up. The materials shall be capable of meeting all of the operational and environmental requirements specified herein. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 4.4.1).

3.2.1 Corrosion resistance. The pickup shall not be affected by drippage of military type fuels, lubricating oils, cleaning solvents, and salt water (see 4.4.1).

3.2.2 Potting. If potting is used within the pickup, the cross-sectional area exposed to the external environment shall be kept to an absolute minimum (see 4.4.1).

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3.3 Design and construction. The pickup design shall be such that (see 4.4.2):

a. The pickup does not utilize any wire smaller than 7.11 millimeter (mm) (#41 gauge) American Wire Gauge (AWG).

b. Unloaded or unstressed wall thickness shall be not less than .508 mm [0.020 inch (in)] (this does not pertain to the thickness of the front face shield on blind ended pickups which is limited only by normal, good design practice).

c. Loaded, stressed or threaded wall thickness (taken from the maximum inside diameter to the root of the external thread) shall be not less than .762 mm (0.030 in).

3.3.1 Envelope. Envelope dimensions for each type of pickup shall be in accordance with the requirements of the applicable drawing (see 4.4.2):

Type	Drawing Number
I	12302067
II	12302350

3.3.2 Weight. The weight of the pickup shall be as follows:

Type	Size
I	.15 kilograms (kg) [.33 pounds (lb)]
II	.16 kg [.35 lb]

3.4 Performance.

3.4.1 Insulation resistance. The insulation resistance between the coil and case shall be not less than 1 megohm at 50 volts (V) root mean square (rms) (see 4.4.3).

3.4.2 Dielectric strength. Dielectric strength shall be sufficient for the pickup to sustain 100 V rms applied between the coil and case (see 4.4.4).

3.4.3 Output voltage. The output voltage shall be in accordance with table I for the applicable type of pickup (see 4.4.5).

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TABLE I. Output voltage.

Pickup type	Engine speed (%)	Output voltage (peak)	Load (ohms)	Surface speed (mmps [ips]) 1/	Frequency (pps) 2/	Test gap mm (in)	Coil resistance (ohms)
Type I	5	0.3	4700	3352 (132)	66	2.54 (.1 in)	85-115
	100	5.0	4700	66700 (2626)	1313	2.54 (.1 in)	85-115
Type II	5	0.4	4700	2261 (89)	45	2.54 (.1 in)	80-460
	54	4.0	4700	24562 (967)	483	2.54 (.1 in)	80-460
	108	7.0	4700	49124 (1934)	967	2.54 (.1 in)	80-460

1/ (mmps [ips]) = millimeters per second (inches per second).
2/ pps = pulses per second.

3.4.4 Phasing. The pickup shall be wired so that pin A is positive with respect to pin B when the pickup tip is approached by a ferrous object (see 4.4.6).

3.5 Environmental conditions.

3.5.1 Temperature. The pickup shall operate when exposed to temperatures of -54 to 204 degrees Celsius (°C) (-65 to 400 degrees Fahrenheit (°F)) (see 4.4.7.1).

3.5.2 Altitude. The pickup shall operate when exposed to altitudes of 396 meters (m) (1300 feet (ft)) below sea level to 3048 m (10,000 ft) above sea level (see 4.4.7.2).

3.5.3 Humidity. The pickup shall evidence no corrosion or other defects that would affect the function or structural strength when exposed to zero to 100 percent (%) relative humidity (see 4.4.7.3).

3.5.4 Salt fog. The pickup shall operate when exposed to a salt-fog (see 4.4.7.4).

3.5.5 Sand and dust. The pickup shall operate after 6 hours exposure to blowing dust as specified in 4.4.7.5.

3.5.6 Fungus. The pickup shall evidence no microbial growth that would adversely affect performance after 28 days exposure to a spore incubation as specified in 4.4.7.6.

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3.5.7 Chemical resistance. The pickup shall meet the specified performance requirements when exposed to fuel vapors, diesel fuel oil, military combat gasoline aviation turbine fluid; solvents and steam cleaning fluid (P-C-437) in operating, nonoperating and transit conditions (see 4.4.7.7).

3.5.8 Vibration. The pickup shall meet the specified performance requirements when exposed to vibration having levels up to 5 gravity units (g) from 5 to 2000 Hertz (Hz) (see 4.4.7.8).

3.5.9 Shock. Pick up shall not be damaged when subjected to the shock specified in 4.4.7.9.

3.5.9.1 Gun firing shock. The pickup shall meet the specified performance requirements without malfunction when exposed to shocks of 100 ± 10 g for 1.5 ± 0.2 millisecond (ms) duration in any direction (see 4.4.7.9.1).

3.5.10 Submergence. The pickup shall operate without short- or long-term damage after being submerged in the fluid (see 3.5.7) at a pressure of 50 pounds per square inch gage (psig) for a period of 2 hours (see 4.4.7.10).

3.5.10.1 Ballistic shock. The pickup shall meet the specified performance requirements without malfunction when exposed to half sine wave shock pulses of 200 ± 20 g in any direction (see 4.4.7.9.2).

3.5.10.2 Basic shock. The pick up shall meet the specified performance requirements without malfunction when exposed to half sine wave shock pulses of 40 g for 18 ms duration in any direction (see 4.4.7.9.3).

3.6 Identification marking. Identification marking shall be in accordance with MIL-STD-130 (see 4.4.2 and 6.2).

3.7 Workmanship. The pickup shall be fabricated and finished in such a manner that criteria of appearance, fit and adherence to specific tolerances shall be observed. Particular attention shall be given to wiring, marking, plating and freedom of parts from burrs and sharp edges (see 4.4.2).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 3.1 and 4.2).
b. Conformance inspections (see 4.3).

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4.2 First article inspection. First article inspection shall be evidenced that all items already produced prior to completion of the first article inspection is similarly deficient unless contrary evidence satisfactory to all items shall be corrected by the contractor. First article samples shall be inspected as specified in table II.

TABLE II. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance	
				Examination	Acceptance tests
Materials and construction	3.2 thru 3.2.2	4.4.1	X		
Defects (see table III)	3.3 thru 3.3.2, 3.6, and 3.7	4.4.2	X	X	
Insulation resistance	3.4.1	4.4.3	X		X
Dielectric strength	3.4.2	4.4.4	X		X
Output voltage	3.4.3	4.4.5	X		X
Phasing	3.4.4	4.4.6			X
Temperature	3.5.1	4.4.7.1	X		
Altitude	3.5.2	4.4.7.2	X		
Humidity	3.5.3	4.4.7.3	X		
Salt fog	3.5.4	4.4.7.4	X		
Sand and dust	3.5.5	4.4.7.5	X		
Fungus	3.5.6	4.4.7.6	X		
Chemical resistance	3.5.7	4.4.7.7	X		
Vibration	3.5.8	4.4.7.8	X		
Gun-firing shock	3.5.9.1	4.4.7.9.1	X		
Ballistic shock	3.5.10.1	4.4.7.9.2	X		
Basic shock	3.5.10.2	4.4.7.9.3	X		
Submergence	3.5.10	4.4.7.10	X		

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4.3 Conformance inspection.

4.3.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ASQC/ANSI Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table II. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.3.2 Examination. Each sample selected with 4.3.1 shall be examined and shall be free from any defect as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Assembly, incomplete (see 3.3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3 and 3.3.1).	SIE 1/
103	Weight not as specified (see 3.3.2).	Scale
104	Workmanship, faulty affecting performance (see 3.7).	Visual
Minor		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3.1).	SIE
202	Marking, improper (see 3.6).	Visual
203	Workmanship, faulty affecting appearance (see 3.7).	Visual

1/ SIE = Standard Inspection Equipment.

4.3.3 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

4.3.4 Acceptance tests. Each pickup shall be subjected to the acceptance tests specified in table II.

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4.4 Methods of inspection.

4.4.1 Materials. Conformance to 3.2 thru 3.2.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.4.2 Defects. Conformance to 3.3 through 3.3.2, 3.6 and 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.4.3 Insulation resistance. To determine conformance to 3.4.1, measure insulation resistance between the coil and case using an applied potential of 50 V rms.

4.4.4 Dielectric strength. To determine conformance to 3.4.2, 100 V rms shall be applied between the coil and the case.

4.4.5 Output voltage. To determine conformance to 3.4.3, measure the output voltage under the conditions specified in table I using a standard test ring.

4.4.6 Phasing. To determine conformance to 3.4.4, the pickup shall be connected to a dc voltmeter with the positive lead connected to pin A and the negative lead connected to pin B. When the tip of the pickup is approached by a magnetic object, the metal needle shall deflect. The connections shall be reversed and the tip again approached with a magnetic object. There shall be no needle deflection.

4.4.7 Environmental tests. The pickup shall be subjected to the environmental tests specified herein. After completion of each environmental test, the performance tests specified in 4.4.3 and 4.4.5 shall be repeated.

4.4.7.1 Temperature. To determine conformance to 3.5.1, the pickup shall be tested as follows:

a. Accelerated aging. Dry the pickup in an air oven and subject it to a temperature of 204°C (400°F) for 168 hours.
b. High temperature. Upon completion of the accelerated aging, subject the pickup to a temperature of 100°C (212°F) for 100 hours.

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c. Room temperature. Upon completion of the high temperature test, subject the pickup to a temperature of 25°C (77°F) for 8 hours.
d. Low temperature. Upon completion of the room temperature test, subject the pickup to a temperature of -54°C (-65°F) for 30 hours.

4.4.7.2 Altitude. To determine conformance to 3.5.2, the pickup shall be tested as follows. Subject the pickup to an atmospheric pressure of 31.19 inches of mercury (396 m (1300 feet) below sea level). Subject the pickup to an atmospheric pressure of 20.01 inches of mercury (3048 m (10,000 feet) above sea level).

4.4.7.3 Humidity. To determine conformance to 3.5.3, the pickup shall be tested as follows. Subject the pickup to an atmosphere of clear water vapor for a period of 15 days under the conditions and cycle as specified in table IV. Immediately after 12 hours of the first test cycle, check the pickup for proper operation with the maximum specified voltage. During the last cycle of the 15-day test, operate the pickup throughout its entire range of operation under each of the ambient conditions specified. At the end of the 15-day test, dry the pickup without forced convection for 12 hours in air at not greater than 21°C (70°F) with not less than 50% relative humidity. At the conclusion of this period, check the insulation resistance and output voltage for conformance to specification requirements. After this, disassemble and inspect the pickup.

TABLE IV. Humidity conditions test cycle.

Time period (hours)	Relative humidity (%)	Ambient temperature °C (°F)
8	95 ± 5	54 ± -15 (129 ± 5.4)
4	Holding chamber moisture content constant	54 ± -15 (129 ± 5.4) gradually down to
8	Holding chamber moisture content constant	21 ± -15 (70 ± 5.4)
4	Up to 95 ± 5, holding chamber moisture content or adding vapor to reach required value	21 ± -15 (70 ± 5.4) uniformly up to 54 ± -15 (129 ± 5.4)

4.4.7.4 Salt fog. To determine conformance to 3.5.4, the pickup shall be tested as follows. Expose the pickup to a salt fog for not less than 48 hours. The fog solution shall be 5% sodium chloride (NaCl) and 95% distilled water by weight. The temperature shall be maintained between 32°C (90°F) and 35°C (95°F). Fog density shall be 3 quarts of solution per 28.31 m^3 (10 ft^3) of chamber volume per 24 hours.

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4.4.7.5 Sand and dust. To determine conformance to 3.5.5, the pickup shall be tested as follows. Subject the pickup to a sand and dust test in accordance with MIL-STD-810, method 510. The pickup need not be operated during the test; however, at the conclusion of the test, and prior to any cleaning of the pickup, check insulation resistance and output voltage for conformance to specification requirements. Criteria for passing the test shall be as defined in MIL-STD-810, method 510.

4.4.7.6 Fungus. To determine conformance to 3.5.6, the pickup shall be tested as follows. Inoculate the pickup with a spore suspension and test in accordance with the requirements of MIL-STD-810, method 508. The temperature shall range from 27°C (80°F) to 29°C (84°F) and the relative humidity from 96 to 100% for a period of 28 days.

4.4.7.7 Chemical resistance. To determine conformance to 3.5.7, the pickup shall be exposed to the chemicals as specified for not less than 48 hours.

4.4.7.8 Vibration. To determine conformance to 3.5.8, the pickup shall be tested as follows. When normally mounted and while powered, subject the pickup to vibrations of 5 g magnitude in each direction along each of 3 mutually perpendicular axes at a logarithmic sweep rate. The total cycling vibration time shall be equivalent to 20 minutes per axis from 5 to 2000 to 5 Hz with each 20- minute test period representing 1609 km (1000 miles) of vehicle operation. Conduct a resonance sweep to identify up to four of the most severe resonances and then accomplish resonance dwells for 1/6 of the total cycling vibration time at each resonance at the specified levels.

4.4.7.9 Shock.

4.4.7.9.1 Gun firing shock. To determine conformance to 3.5.9.1, the pickup shall be tested as follows. When normally mounted and while powered, subject the pickup to 3 shocks of 100 ± 10 g for 1.5 ± 0.2 ms along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.9.2 Ballistic shock. To determine conformance to 3.5.10.1, the pickup shall be tested as follows. When normally mounted and while powered, subject the pickup to 3 half sine wave shock pulses of 200 ± 20 g along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.9.3 Basic shock. To determine conformance to 3.5.10.2, the pickup shall be tested as follows. When normally mounted and while powered, subject the pickup to 3 half sine wave shock pulses of 40 g for 18 ms along each of 3 mutually perpendicular axes (a total of 18 shocks).

4.4.7.10 Submergence. To determine conformance to 3.5.10, the pickup shall be completely submerged in the fluids specified in 3.5.7. The duration of submergence shall be 2 hours to insure total submergence of the pickup. The pressure of the fluid shall be maintained at 345 kPa (50 psig).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The pickup covered by this specification is intended to be used to generate signals reflecting the two rotor speeds within the AGT 1500 gas turbine engine of the M1 tank.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. When first article is required (see 3. 1).
d. Identification and marking if other than as specified (see 3.6).
e. If inspection conditions shall be other than as specified (see 4.2).
f. Selection of applicable packaging requirements (see 5.1).

6.3 Subject term (key word) listings.

AGT 1500 Gas turbine engine
Magnetic speed pickup
Pickup, magnetic
Rotor speeds
Signals

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Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-62413A(AT)	951204

3. DOCUMENT TITLE
Speed Pickup, Magnetic (Metric)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
	(810) 574-8745	DSN 786-8745
c. ADDRESS (Include Zip Code) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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